Regulatory Relevance and Prospective Validation

Working section for *From Formal Authority to Practical Human Control*

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The institutional gap

Governance instruments can assign oversight duties without producing evidence that a person could influence a particular decision. A policy may require a reviewer, a notice, an audit, or a stop control. The institutional claim becomes stronger only when records show that the person received usable information, understood the system and situation, had authority and time to act, exercised judgment, and changed execution.

The practical-human-control method tests that connection. It does not issue a legal-compliance or certification finding.

Relationship to selected instruments

Instrument	Reviewed function	Research use
EU AI Act, Article 14	Understanding limits, automation-bias awareness, interpretation, override, reversal, intervention, and stopping	Test selected capabilities; record information delivery, opportunity, exercised judgment, and propagation separately
EU AI Act, Article 57	Controlled testing and validation under regulatory supervision	Capture every stage prospectively before the record becomes incomplete
NIST AI RMF Core	Documented roles, assessed operator proficiency, and defined oversight processes	Connect organization-level arrangements to event-level evidence requests
ISO/IEC 42001 public overview	AI management-system governance, risk, traceability, transparency, reliability, and improvement	Place an event-level evidence layer inside an organizational system; no conformity or certification claim

The reviewed instruments describe capabilities and processes that institutions should establish. Their existence alone does not show that those arrangements had practical force in one event. The six-stage method asks what records show that assigned oversight became timely, informed, exercised, and propagated control.

Why a sandbox is the next empirical step

An EU AI Act regulatory sandbox can test the method with evidence collected by design. Before a run, the study should freeze the system and decision boundary, six required stages, timing definitions, intervention controls, evidence states, stopping rule, and output schema.

During the run, the study should preserve the exact information shown to the overseer, system version and limits, training and role records, complete timestamps, the contemporaneous judgment or intervention, and proof that the intervention reached downstream execution.

A solo rehearsal can test internal executability. A blinded second assessor is required for reliability. Field validity also requires target-system and task sampling and an external criterion. Work involving participants or protected operational records requires the applicable ethics determination and data authority before collection.

Prospective evidence plan

Stage	Required record	Prospective test
Information access	Timestamped interface state and delivered evidence	Verify what the overseer could see before irreversible commitment
Comprehension capacity	Training basis, limitation disclosure, and task-specific comprehension check	Test interpretation of output, uncertainty, and alternatives under study conditions
Intervention authority	Role delegation and tested controls	Verify that approve, reject, modify, stop, and escalation rights function
Intervention feasibility	Workload, latency, staffing, and complete timing inputs	Calculate the proposed timing margin and record channel failures
Exercised judgment	Contemporaneous decision, challenge, or intervention record	Distinguish active judgment from automatic confirmation
Execution propagation	Linked downstream state or action record	Verify that the intervention changed execution or a binding obligation

The first study should contain a functioning control path, a path with one prespecified operational constraint, and a missing-record path. The third condition checks whether the method returns unresolved when the record cannot decide. These conditions test procedure behavior and supply no prevalence estimate.

United States applied-policy sites

Setting	Reviewed duty or right	Current boundary
New York City	Bias audit, public audit information, and notice for covered employment tools	No inference that a named person could alter an individual decision
California	Access and opt-out rights for covered ADMT uses	Compliance begins 1 January 2027; no event-level case assessed
Colorado	Consequential-decision framework effective 1 January 2027	Detailed mapping waits for final implementing rules
Illinois and Chicago	Employment discrimination prohibition and notice	Chicago remains within the state entry until a municipal source is selected

Claim boundary

The crosswalk proposes research uses for selected instruments. It does not interpret every legal obligation, establish compliance, certify an AI management system, provide legal advice, or show that the method improves institutional outcomes. The eight source observations require accountable author review before manuscript inclusion.

Kevin Baum's private message identifies a possible audience and application setting. It is excluded from the evidence base and supplies no validation or peer-review result.

Official sources

European Union: Regulation (EU) 2024/1689, Articles 14 and 57
National Institute of Standards and Technology: AI RMF Core
ISO and IEC: ISO/IEC 42001:2023 public overview